

Wan Elaina Farah binti Wan Shazril Hafiz

Aspiring Data Engineer

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Nationality: Malaysia | Languages: Bahasa Melayu (Fluent), English (Fluent), German (Basic)

PROFESSIONAL SUMMARY

A Diploma in Computer Science student at Universiti Teknologi Malaysia (UTM) with a strong foundation in low-level programming, relational database systems, and structured problem-solving. Highly motivated Data Engineering enthusiast with hands-on academic exposure to relational data modelling, query optimisation, and the cleaning of structured datasets using Python and flat text formats. Combines a highly logical approach to software development with creative problem-solving skills and practical, large-scale leadership experience coordinating high-volume administrative operations.

AREAS OF EXPERTISE & SKILLS

Technical Skills

- **Programming:** C++, HTML, CSS, Python, JavaScript, PHP
- **Databases & Tools:** MySQL, phpMyAdmin, XAMPP / WAMP
- **Data Handling:** Cleaning, filtering, and importing flat data (CSVs) using Pandas library
- **Environments:** Visual Studio Code, Git, GitHub, Git Bash, Dev-C++, Excel

Practical Skills

- **Creative Problem-Solving:** Building custom personal productivity suites and visual web layouts
- **Operational Leadership:** Coordinating large volunteer/staff teams (up to 60 people)
- **Data Integrity & Auditing:** Translating physical manual records into clean spreadsheet reports
- **Team Collaboration:** Clear documentation, structured task delegation, and conflict resolution

TECHNICAL PROJECT EXPERIENCE

Colour-Coded Access Control Panel (Digital Logic Project)

UTM KL

Hardware Circuit Logic Simulation & Optimisation via DEEDS

- Designed a room entry validation panel that reads authorised identity card numbers as digital keys using a 16-to-1 multiplexer.
- Mapped truth tables and Karnaugh maps (K-Maps) to optimise a network of NOT and AND gates, ensuring only one specific indicator light triggers at any time to prevent circuit validation errors.
- Incorporated sequential logic elements, utilising D-flip-flops for clean edge detection and a BCD counter to seamlessly track and output total valid entries onto a 7-segment display.

E-RegiCat Attendance & Registration System

Academic Project

Web-Based Student Event Management Application

- Architected custom relational tables inside MySQL, manually building primary-foreign key linkages and correct structural cardinality lines.

- Authored manual, explicit SQL queries wrapped in PHP to retrieve and update records dynamically, avoiding data redundancies.
- Replaced traditional manual paper handling with a structured web portal, significantly boosting coordination efficiency.

Technologies used: PHP, MySQL, HTML, CSS, JavaScript, phpMyAdmin

Hospital Management & Appointment System (Prototype)

Database Design Project

Relational Database Workspace Simulation

- Created a normal form relational database containing distinct, connected tables for Patient Records, Appointment Scheduling, and Medical History.
- Wrote granular, manual SQL lookup queries to simulate complex cross-table data requests, modelling ways to reduce hospital counter waiting lines.

Personal Development Tools Suite & Web Scrapbook

Personal Technical Sandbox

Custom Programmed Utility Packages & Neocities Digital Archive

- Coded standard programmatic tool scripts, including a task tracker, finance tracker, mathematical converters, and BMI calculators to test input-validation conditions.
- Built a structured personal multi-page "scrapbook" website on Neocities to test custom file pathways, directory layouts, and direct CSS styles.

EDUCATION

Universiti Teknologi Malaysia (UTM KL)

Expected Graduation: 2028

Diploma in Computer Science (DSPD)

- **Core Modules Include:** Database Systems, Data Structures & Algorithms, Digital Logic, System Development, Computer Science Programming, and Professional Ethics.
- **Data Structures Focus:** Coded custom low-level memory structure configurations, including Linked Lists, Stacks, Queues, and Hierarchical Trees using modular source and header file formats.

SMK Puncak Alam, Selangor

Graduated

Secondary Education — STEM Stream

- **SPM Achievements:** Obtained 1A+, 3A, 2A- (Strongest performance trends in Computer Science, Mathematics, English, Physics, and History).

LEADERSHIP & PRACTICAL EXPERIENCE

School Cooperative Secretary / Operations Member

Cooperative Society

Scale: Serving 500+ Campus Students & Teachers

- **Weekly Reporting Pipelines:** Maintained on-the-ground sales data via physical daily logboxes, executing weekly data aggregation projects to combine, clean, and organise total earnings into master Microsoft Excel administrative reports.

- **Operational Coordination:** Worked collaboratively with a team of 60 rotating student staff members, applying structured, fair delegation across shifts, counter service, bookkeeping, and asset management.

WORK EXPERIENCE

Administrative Assistant / Cashier (Part-Time)

Indoor Playground Facility

- Managed visitor reception records and customer intakes, keeping straightforward, organised administrative front-desk log sheets.
- Handled point-of-sale transactions carefully, tracking payment logs to reconcile terminal balances with zero errors.
- Provided responsive, friendly communication to coordinate visitor crowds safely during busy weekend peaks.

Family Business Support

Maajun Nenda (Herbal Products)

- Followed exact product recipes to measure raw natural ingredients precisely, preserving consistency across production batches.
- Packed incoming finished inventory cleanly based on specific target weights and item sorting matrices.

Sales Assistant / Clerk

Bata Retail

- Arranged store floor alignments, cross-checked incoming backend supply boxes, and handled cash register operations.

CERTIFICATIONS & ACHIEVEMENTS

- **FreeCodeCamp:** Data Analysis with Python Certification
- **Google Career Certificates:** Data Analytics Professional Certificate (In progress)
- **Cisco Networking Academy:** Network Infrastructure Fundamentals (In progress)
- **Anugerah Akademik:** Earned top 10 performance recognition within the academic class stream tracking.
- **Best Secretary Award:** School-level commendation for clear administrative document layouts and record management.
- **Co-Curricular Engagements:** Programming meetups (participation-based), Essay Competitions, Scrabble, Chess, and School Debate Team.

CAREER VECTOR TARGETS

- **Core Aspirations:** Data Engineering, Database Architecture, Systems Development.
- **Industries of Interest:** Healthcare Information Infrastructure, Biotech Systems, Supply Chain Analytics, and Aviation Data Logistics.